

Appendix 30.B

3G Cellular – WCDMA/UMTS

This Appendix is a short summary of the *Wideband Code Division Multiple Access* (WCDMA) standard for third generation (3G) cellular telephony. Just like the Global System for Mobile communications (GSM) described in Appendix 30.A, WCDMA is a somewhat outdated technology, but still used in a considerable percentage of networks throughout the world. This standard is also part of a group of standards that are known as the *Universal Mobile Telecommunication System* (UMTS), *Third Generation Partnership Project* (3GPP), and *International Mobile Telecommunications* (IMT-2000). Note that this Appendix only describes the “basic” WCDMA system; for the high-speed packet mode HSPA see the references in “Further Reading”.

30.B.1 Historical Overview

In this section, we will review the history of this standard and show its relationship to other third-generation standards.

The success of second-generation cellphones, especially GSM, motivated the development of a successor system. The International Telecommunications Union (ITU) announced goals for a standard for a 3G system, called IMT-2000:

- better spectral efficiency;
- higher peak data rates – namely, up to 2 Mbit/s indoor and 384 kbit/s outdoor;
- supporting multimedia applications, meaning the transmission of voice, arbitrary data, text, pictures, audio and video, which requires increased flexibility in the choice of data rates;
- backward compatibility to second-generation systems.

Europe started research activities toward this goal in the early 1990s, and finally selected in January 1998 two drafts as their candidates: broadband CDMA – also known as *Frequency Division Duplexing* (FDD) mode – which was intended as the basic system, and T/CDMA with Joint Detection-Time/Code Division Multiple Access (JD-TCDMA) – also known as Time Division Duplexing (TDD) mode – which was to support high-data-rate applications. The FDD mode and TDD mode were often subsumed under the name WCDMA. FDD mode is the more important part, and will be at the center of our attention in this Appendix. WCDMA is mostly used as an abbreviation for radio access technology, while the expression UMTS refers to the complete system, including the Core Network (CN).

These two systems were then included as the European proposal for the IMT-2000 family. A Japanese proposal for a WCDMA system was fairly similar to the European proposal, and thus merged. Still, it was not possible to reach agreement for a single 3G system within the ITU. Besides the Japanese/European proposal, the CDMA 2000 proposal, an evolution of the CDMA-based IS-95 2G system, had strong support. Several other proposals were advocated as well. In order to avoid a deadlock, the ITU decided to declare all of these proposals to be valid “modes” of IMT-2000.

The further development of the merged Japanese/European standard was done by the 3GPP industry organization, whose members include ETSI, ARIB (Association of Radio Industries and Businesses, the Japanese standardization organization) and several other members. The original specifications are also known as “Release 99.” Subsequent improvements were included in later releases – in particular High-Speed Packet Access (HSPA), which was first realized for the downlink, and then also the uplink. While there were advantages to having a broad base of companies contributing to the specifications, 3GPP faced some serious problems: the size of the specifications grew beyond reasonable limits, and became so complicated that it hampered implementation. The resulting high development costs and delays in time to market almost led to the death of the system, and it started to take off only around 2005. The exception was Japan, where implementation proceeded much faster than elsewhere. It is worth noting that Japan did not implement the full

UMTS specifications, but rather a simplified system called FOMA. Development in Japan was also helped by the fact that second-generation Japanese Digital Cellular (JDC) reached its capacity limits earlier and the alternative Personal Communication Systems (PCSs) (compare with the DECT in Appendix 30.C) could not meet demand completely.

UMTS development was also hampered by regulatory developments. The first UMTS-related licenses were granted first in 1999 in Finland and the other European countries followed in 2000. Particularly, the bidding process for the licenses in the United Kingdom and Germany received a lot of public attention, as several billion Euros were finally paid for each license. According to the regulatory authorities, the first UMTS networks should have started operating in 2002. However, the real start of operation was delayed due to technical problems and the fact that the market for mobile high-data-rate applications had to be developed first. Only by the end of 2003 did the major mobile providers in Europe start to offer subscriptions to UMTS and the appropriate devices. But by 2009 these problems had been overcome, and UMTS turned out to be extremely successful all over the world. However, after 2009, the emergence of LTE led to a decline in subscriber numbers, which is anticipated to decrease over the next years, as 3G networks are turned off and replaced with LTE or 5G. Still, just like for GSM, there is even currently (2022) a sizeable user base (roughly 1 billion subscribers) throughout the world left, and knowledge of WCDMA can thus be useful.

The remainder of this Appendix describes mainly the physical layer of WCDMA/UMTS. Many of the networking functions are similar to GSM, and we simply refer the reader to Appendix 30.A for these.

30.B.2 System Overview

30.B.2.1 Physical-Layer Overview

We first summarize the physical layer of WCDMA – i.e., communications between the User Equipment (UE) and the Base Station (BS) via the air interface. The UMTS standard uses a number of unusual abbreviations. For example, the BS is called *Node-B* in the UMTS standard. In order to stay consistent with the notation in the remainder in this book, we will stick to BS.

The WCDMA air interface uses CDMA for distinguishing between different users, and also between users and some control channels. We distinguish between spreading codes that are responsible for bandwidth expansion of the signals, and scrambling codes that are mainly used for distinguishing signals from different UEs and/or BSs. In addition, WCDMA also shows a timeslot structure: the time axis is divided into units of 10 ms, each of which is subdivided into slots of 0.67 ms. Depending on the location within a timeslot, a symbol might have different meanings.

WCDMA uses a number of logical channels for data and control information, which will be discussed in Section 30.B.4. They are then mapped to physical channels – i.e., channels that are distinguished by spreading codes, scrambling codes, and positions within the timeslot.

30.B.2.2 Network Structure

To discuss how a mobile network provider may introduce UMTS, we have to distinguish between the *UE*, the *Radio Access Network* (RAN), and the CN. The UE and the *UMTS Terrestrial Radio Access Network* (UTRAN) communicate with each other via the air interface, as discussed in the previous section. The UTRAN consists of multiple *Radio Network Subsystems* (RNSs), each of which contains several *Radio Network Controllers* (RNCs), each of which controls one or several BSs (Node-Bs).

The CN connects the different RNSs with each other and other networks, like the internet. The CN can be based on an upgraded GSM CN or might be implemented as a completely new Internet Protocol (IP)-based network. Details about the different functional units of a GSM CN (mobile switching center, home location register, etc.) can be found in Appendix 30.A. The network functionalities for packet data are similar to the ones of the General Packet Radio Service (GPRS) (see Appendix 30.A.8 and Appendix 24.B of the second edition).

Another way of looking at the UMTS architecture is to organize it in two domains:

1. *UE domain*, which consists of:
 - *User Service Identity Module* (USIM).
 - *Mobile Equipment* (ME) consisting of:

- *Terminal Equipment* (TE);
 - *Terminal Adapter* (TA);
 - *Mobile Termination* (MT).
2. *Infrastructure domain*, which consists of:
- The access network domain consisting of:
 - UTRAN.
 - The CN domain consisting of:
 - *Inter Working Unit* (IWU);
 - serving network;
 - transit network;
 - home network;
 - application network.

The physical properties of a link between two network entities and the signals transmitted via this link together with the functions of these signals are collectively referred to as an *interface*. Usually, an interface is standardized. Figure 30.B.1 shows the relevant interfaces in UMTS.

30.B.2.3 Hierarchical Cellular Structure

UMTS is intended to achieve global availability and thereby enable roaming worldwide. Therefore, the coverage area in UMTS is divided hierarchically into layers. The higher layers cover a bigger area than the lower layers. The highest layer achieves worldwide coverage by using satellites.¹ The lower layers are the macrolayer, microlayer, and picolayer. They constitute the *UTRAN*. Each layer consists of several cells. The lower the layer the smaller the cells. Thus, the macrolayer is responsible for nationwide coverage with macrocells. Microcells are used for additional coverage in the urban environment and picocells are employed in buildings or “hotspots” such as airports or train stations. This concept is known and has been discussed and partly implemented for a while. However, UMTS was supposed to cover all aspects of this concept worldwide and with the initial rollout. In practice, however, it has been rather different: in the initial phase, only the big cities were covered with a few cells, and large-area coverage was achieved by dual-mode devices that can communicate with either WCDMA or EDGE/GPRS/GSM networks. Only in later buildout stages (often after 2010) was coverage of all areas with WCDMA achieved.

30.B.2.4 Data Rates and Service Classes

The maximal data rate and the highest supported user velocity are different for each hierarchy layer. The macrolayer supports at least 144 kbit/s at speeds up to 500 km/h. In the microlayer data rates of 384 kbit/s at maximal speeds of 120 km/h are achievable. In picocells, maximal user speeds of 10 km/h and maximal data rates of 2 Mbit/s are supported. Figure 30.B.2 compares data rates and maximal user velocity with other cellular (GSM) or wireless standards.

Maximal Bit Error Rates (BERs) and transmission delays are grouped into sets out of which the user may choose the following:

- *Conversational*: this class is mainly intended for speech services, similar to GSM. The delays for this type of service should be on the order of 100 ms or less; larger values are experienced as unpleasant interruptions by users. BERs should be on the order of 10^{-4} or less.
- *Streaming*: audio- and videostreaming are viewed as one important application of WCDMA. Larger delays (in excess of 100 ms) can be tolerated, as the receiver typically buffers several seconds of streaming material (see Chapter 20). BERs are typically smaller, as noise in the audio (music) signal is often considered to be more irksome than in a voice (telephone) conversation.
- *Interactive*: this category encompasses applications where the user requests data from a remote appliance. The most important category is Web browsing, but also database retrievals and interactive computer games

¹ This was not widely implemented. Satellite systems for cellular phones are now considered as a possibility for advanced 5G systems.

fall into this category. Also for this category, there are upper limits to tolerable delay – the time between choosing a certain website and its actual appearance on the screen should not exceed a few seconds. BERs have to be lower, typically 10^{-6} or less.

- *Background class*: this category encompasses services where transmission delays are not critical. These services encompass email, Short Message Service (SMS), etc.

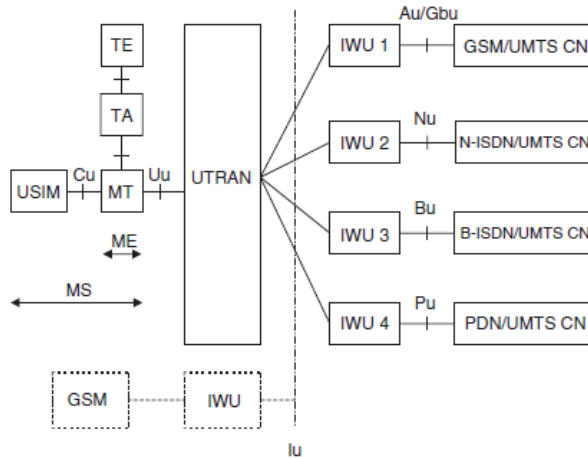


Figure 30.B.1 Interfaces in Universal Mobile Telecommunications System. MS: mobile station, i.e., UE.

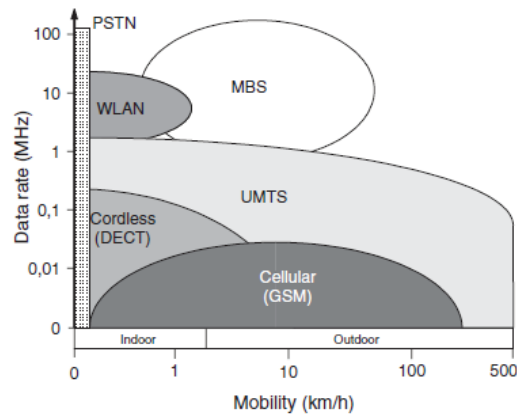


Figure 30.B.2 Data rates versus mobility for Universal Mobile Telecommunications System, Global System for Mobile communications, Digital Enhanced Cordless Telephone, wireless local area network mobile broadband systems and landline networks.

30.B.3 Air Interface

30.B.3.1 Frequency Bands and Duplexing

In most of the world, UMTS utilizes frequencies in ranges from 1,900 MHz to 2,025 MHz and from 2,110 MHz to 2,200 MHz (see Figure 30.B.3). Within these bands there is a dedicated subband for the *Mobile Satellite*

*Service (MSS).*² The MSS uplink uses the band from 1,980 MHz to 2,010 MHz and the MSS downlink the band from 2,170 MHz to 2,200 MHz. The remaining parts of bands are split for the two modes of terrestrial operation, UMTS-TDD and UMTS-FDD. UMTS-FDD uses the band from 1,920 MHz to 1,980 MHz for the uplink and the band from 2,110 MHz to 2,170 MHz for the downlink. As the name indicates, TDD mode does not distinguish uplink and downlink by using different carrier frequencies but rather by accessing different timeslots on the same carrier. Therefore, this mode does not require symmetric frequency bands. It can simply use all of the remaining frequencies.

Some regions of the world had different default bands (e.g., USA), and over time a number of other bands were assigned to UMTS and other 3G services as well. Since 2010, many of those bands have been re-assigned (by the frequency regulators and/or the operators) to 4G and 5G.

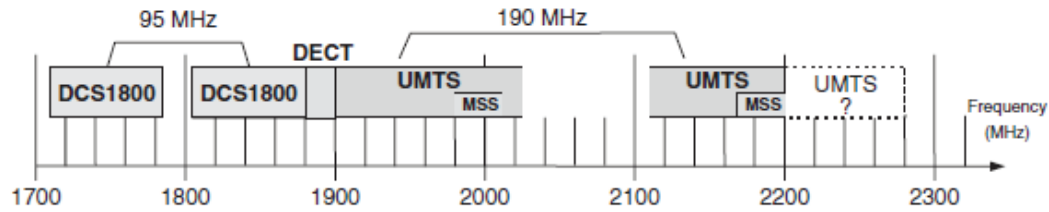


Figure 30.B.3 Frequency bands originally allocated to Universal Mobile Telecommunications System.

30.B.3.2 Time Domain Duplexing and Frequency Domain Duplexing Modes

FDD operating mode is intended for use in macrocells and microcells, whereas TDD operating mode is intended for use in picocells. In the TDD mode it is more difficult to handle large propagation delays between the UE and BS, as transmitting and receiving timeslots might overlap. However, TDD has the advantage that big asymmetries in throughput on the downlink and uplink can be easily supported. This is particularly important for applications like Web browsing for which the UE receives much more information than it transmits. For medium access, TDD mode employs JD-TCDMA and FDD mode WCDMA. We will not, however, consider the TDD mode further in this Appendix.

30.B.3.3 Radio-Frequency-Related Aspects

Power Classes and Receiver Sensitivity

UEs: these are divided into four classes, according to their transmit power. The maximal powers are 33, 27, 24, and 21 dBm, corresponding to power classes 1, 2, 3, and 4. Power is measured before the antenna. Thus, the antenna characteristic does not have an impact on these limits. The receiver of the UE has to be so sensitive that given a received signal power of -117 dBm per 3.84-MHz channel a BER of 10^{-3} is still achievable with a 12.2-kbit/s data rate. Note that this specification includes the effects of forward error correction.

BS: there are no transmission powers specified for the BS. However, typical values for the transmit power are in the 10–40 W range. The receiver in the BS has to be so sensitive that given a received signal power of -121 dBm a BER of 10^{-3} for a 12.2-kbit/s is still feasible. Since noise power in a bandwidth of 12.2 kHz is -133 dBm, this is a quite challenging task.

The specifications also prescribe the resistance of the receiver to blocking – i.e., being able to function even in the presence of strong interferers. Processing gain can be used to decrease the effect of interference. However, if the Radio Frequency (RF) elements of the receiver have a limited dynamic range, then the interferer might drive the receive amplifier into saturation, resulting in a signal from which the desired part can no longer be recovered. Similarly, limits to intermodulation and other RF effects are specified [Richardson 2005].

² For a detailed description of bands for fourth generation (4G) systems, see Chapter 31.

Frequency Bands

The regular nominal intercarrier spacing is 5 MHz. However, the network provider might select the carrier distance to be any multiple of 200 kHz. Therefore, carriers are referred to using the *UTRA Absolute Radio Frequency Channel Number* (UARFCN) which refers to multiples of 200 kHz. The frequency deviation of the local oscillators of the UE is limited to 10^{-7} (0.1 parts per million), which equals roughly 200 Hz.

From a purely theoretical point of view, emissions outside the assigned 5-MHz band should be zero. The basis pulse for modulation is a raised cosine pulse with a roll-off factor of $\alpha = 0.22$. Given a chip rate of 3.84 Mcchip/s the signal bandwidth is $(1 + \alpha)/T_C = 4.7$ MHz.³ However, due to nonideal filter implementation, emissions outside this band do appear in practice. These emissions include both out-of-band emissions (defined as emissions a distance of 2.5 MHz to 12.5 MHz from the center frequency) and spurious emissions, which refer to emissions farther away from the center frequency.

Out-of-Band Emissions

There is a set of limitations for out-of-band emissions at a distance of 2.5 MHz to 12.5 MHz:

- **Spectrum emission mask:** Figure 30.B.4 shows the emission mask for a BS. For a given distance Δf to the center frequency, there is a limit for the maximum emitted power within a 30-kHz and 1-MHz bandwidth, respectively.
- **Adjacent Channel Leakage Ratio (ACLR):** this is a measure of how much power leaks from the desired band into the adjacent band. This ratio should be better than 45 dB and 50 dB for the first or second adjacent channel (5-MHz or 10-MHz carrier distance), respectively.

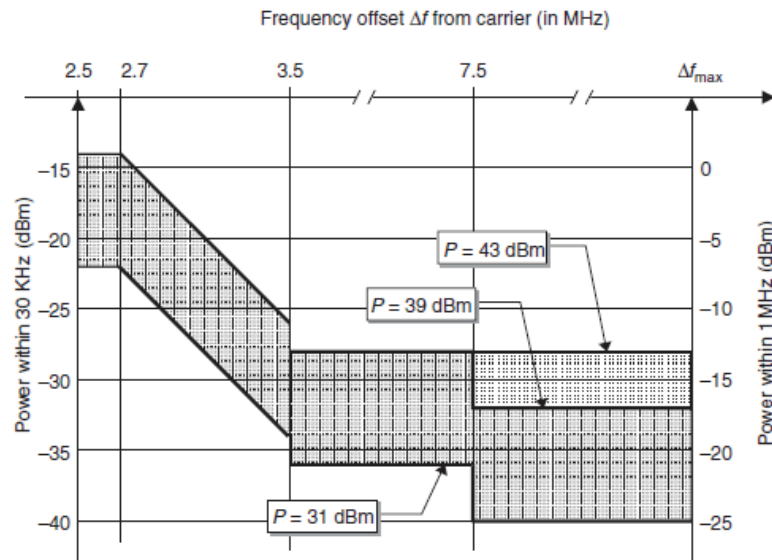


Figure 30.B.4 Spectrum mask for a base station in Wideband Code Division Multiple Access.

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³ The mismatch between signal bandwidth and nominal carrier distance can only be explained from a historic perspective. Originally, a chip rate of 4.096 Mcchip/s was planned. With $\alpha = 0.22$ this results in a signal bandwidth of 5 MHz.

Spurious Emissions

Spurious emissions are emissions far away from the used channel ($\Delta f > 12.5$ MHz). They are due to harmonics emissions, intermodulation products, etc. The limits for these emissions are rather soft – e.g., -30 dBm for a 1-MHz bandwidth. However, for frequencies which are used by other mobile systems, such as GSM, DECT, or UMTS Terrestrial Radio Access (UTRA-TDD), they are more severe. Especially when a UMTS BS is colocated with a GSM BS, emission limits of -98 dBm/100 kHz might apply, thus ensuring that UMTS signals do not interfere with GSM signals.

30.B.4 Physical and Logical Channels

30.B.4.1 Logical Channels

Similar to GSM, we distinguish between different logical channels in UMTS, which are mapped to physical channels. The logical channels in UMTS are sometimes referred to as *transport channels*. There are two kinds of transport channels: *common transport channels* and *Dedicated (transport) CHannels* (DCHs).

Common Channels

Common channels are relevant to all or at least a group of UEs in a cell. Thus, all of them receive the information transmitted on these channels in the downlink and may access the channels in the uplink. There are different kinds of common channels:

- *Broadcast CHannel* (BCH): the BCH is only found in the downlink. Both cell-specific and network-specific information is transmitted on it. For example, the BS uses this channel to inform all UEs in the cell about free access codes and available access channels. This channel has to be transmitted with relatively high power, as all UEs within the cell have to be able to receive it. Thus, on this channel neither power control nor smart antennas are implemented.
- *Paging CHannel* (PCH): this is also a channel that can be found only in the downlink. It is used to tell a UE about an incoming call. Since attenuation of the channel to the UE, as well as the location of the UE, is not known the PCH is transmitted with high power and without employing smart antennas. Depending on whether the current cell of the UE is known, paging information is either transmitted in only one cell or several cells.
- *Random Access CHannel* (RACH): the RACH is only used in the uplink. The UE uses it to initialize a connection to the BS. It can employ open-loop power control, but no smart antennas, as the BS must be able to receive signals on the RACH from every UE within the cell. As it is a *random* access channel, collisions might occur. Therefore, the structure of bursts in the RACH are different from that of other channels – this is described below in more detail.
- *Forward Access CHannel* (FACH): the FACH is used to transmit control information to a specific UE. However, as the FACH is a common channel and therefore received by more than one UE, explicit addressing of the desired UE is required (*in-band identification*, UE-ID at the beginning of the packet). In contrast with this, a dedicated control channel has an implicit addressing of the desired UE: the UE is specified by the carrier and the spreading code used. The FACH can also transmit short user information packets. It can employ smart antennas, as information is transmitted to one specific, localized UE.
- *Common Packet CHannel* (CPCH): the CPCH is an uplink channel, and can be interpreted as the counterpart of the FACH. It can transmit both control and information packets. If the FACH and the CPCH are used together closed-loop power control is possible.
- *Downlink Shared CHannel* (DSCH): the DSCH is a downlink channel similar to the FACH. It sends mainly control data, but also some traffic data, to multiple UEs. Explicit addressing has to be used on this channel, as the same CDMA code is used for all UEs. The reason for using the same code for multiple UEs lies in the limited amount of *short* spreading codes (see next section). Under normal circumstances, one spreading code would be permanently reserved for one UE, even if the traffic is bursty. The cell would thus quickly run out of codes (remember that there are very few codes that can be used for high-data-rate traffic). On the DSCH one short code is used for several UEs and data are multiplexed in time. The DSCH supports fast

power control, use of smart antennas, and rate adaptation from transmission frame to transmission frame. Note that the DSCH uses a different approach in HSDPA – namely, the use of multiple short codes simultaneously.

Dedicated Channels

Dedicated channels are present in both the uplink and the downlink. They are used to transmit both higher layer signaling and actual user data. As the position of the UE is known when a dedicated channel is in use, smart antennas, as well as fast power control and adaptation of the data rate on a frame-by-frame basis, can be used:

- *Dedicated (transport) channel (DCH)*: this is the only type of dedicated logical channel. UE addressing is inherent as for each UE a unique spreading code is used.

30.B.4.2 Physical Channels

WCDMA transmits control and user data on the same logical channel (in contrast to GSM) – namely, the DCH. However, for physical channels we distinguish between channels for control and user information. Combined transmission of both is called the *Coded Composite Traffic Channel* (CCTrCH). Note that there are also some physical channels that are not related to a specific logical channel.

Uplink

In the uplink we find dedicated control and user data channels, which are transmitted simultaneously via I- and Q-code multiplexing (see Section 30.B.6):

- Pilot signals, *Transmit Power Control* (TPC), and *Feed Back Information* (FBI) are transmitted via the *Dedicated Physical Control CHannel* (DPCCH). Furthermore, the *Transport Format Combination Indicator* (TFCI) may be transmitted via the DPCCH (see also Section 30.B.5.2). The TFCI contains the instantaneous parameter of all data transport channels that are multiplexed on the DPDCH (see below). The spreading factor for the DPCCH is constant – namely, 256. Thus, ten control information bits are transmitted in each slot.
- The actual user data is transmitted via the *Dedicated Physical Data CHannel* (DPDCH). Spreading factors between 4 and 256 can be used. The DPDCH and DPCCH are transmitted at the same time and on the same carrier by using the I- and Q-branch, respectively.
- The *RACH* is not only a logical but also a physical channel (Physical RACH or PRACH). A *slotted ALOHA* approach is used for medium access (see Sec. 18.4.1). The burst structure of the RACH is completely different from that of the dedicated channel. Data packets may be transmitted via the PRACH, too. Packets can be transmitted either in pure PRACH mode, meaning that PRACH bursts as described in Section 30.B.6 are used, or in the *Uplink Common Packet CHannel* (UCPCH). The UCPCH is an extension of the PRACH. It is used in combination with a DPCCH in the downlink. Therefore, fast power control is possible.
- The *Physical Common Packet CHannel* (PCPCH) has a similar burst structure to the PRACH. Information is transmitted after a preamble. Initially, several access preambles are transmitted with ascending transmission power until the BS receives the necessary signal strength. After the BS acknowledges reception, another preamble is transmitted to detect eventual collisions with packets from other UEs that try to access the PCPCH at the same time. Before user data are transmitted, a power control preamble of length 0 or 8 timeslots can be transmitted. Only then are the actual data transmitted; the length of this transmission period is a multiple of the frame duration – i.e., 10 ms.

Downlink

Of course, the dedicated data and control channels, DPDCH and DPCCH, can be found in the downlink, too. However, they are multiplexed in a different manner, which is discussed in Section 30.B.6. The frame and slot timing are shown in Figure 30.B.5.

Furthermore, the downlink features the following common control channels:

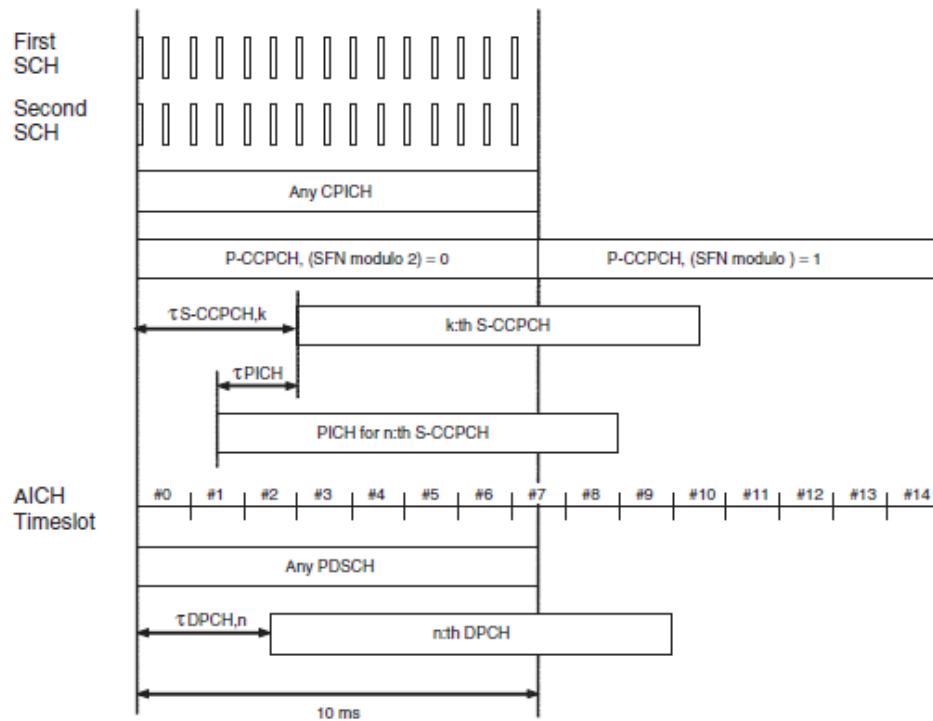


Figure 30.B.5 Frame and slot timing of downlink physical channels.

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- *Primary Common Control Physical CHannel (P-CCPCH)*: bursts transmitted via the CCPCHs are similar to those of the DPCCH. However, there is no power control associated with the CCPCH and therefore the power control bits do not have to be transmitted. The P-CCPCH has an idle period in each frame; this idle period is 256 chips long. The P-CCPCH carries the broadcast channel, and it is thus critical that it can be demodulated by all UEs in the cell. Thus, it uses a constant, high spreading factor – namely, 256.
- *Secondary Common Control Physical CHannel (S-CCPCH)*: the main difference between the primary and the secondary CCPCH is that the data rate and spreading factor are fixed in the former, whereas they are variable in the latter. The S-CCPCH carries the FACH and the PCH.
- *Synchronization CHannel (SCH)*: the SCH does not relate to any logical channel. Its function is explained in Section 30.B.7. The SCH is multiplexed with the P-CCPCH onto one timeslot: it sends 256 chips for synchronization during the above-mentioned idle period of a P-CCPCH burst.
- *Common Pilot CHannel (CPICH)*: this channel has a constant spreading factor of 256. The CPICH consists of a primary and a secondary CPICH. The primary CPICH serves as a reference for phase and amplitude for common channels in the entire cell. The primary CPICH is transmitted all over the cell and the secondary might be transmitted in selected directions. The primary and secondary pilot channels differ in the spreading and scrambling codes used: the primary CPICH always uses the primary scrambling code and a fixed channelization code, so that there is only one such code per cell.
CPICHs are particularly important for establishing a connection as during this period the pilots of the dedicated channels are not available. Furthermore, pilot channels provide an indication of signal strength at

the UE and are therefore important for handover procedures. The cell size of a BS can be varied by varying the transmit power of pilot channels. By reducing transmit power the area over which the BS provides the strongest signal is decreased. This reduces the traffic load of a BS.

- Apart from these channels, the downlink further features the *Physical Downlink Shared CHannel* (PDSCH), which carries the DSCH, the *Acquisition Indication CHannel* (AICH), which provides feedback about whether synchronization was successful or not, and the *Page Indication CHannel* (PICH), which supports paging.

Matching of Logical and Physical Channels

Figure 30.B.6 illustrates how logical channels, also referred to as transport channels, are matched to physical channels. Details regarding frame and slot timing can be found in the standard.

30.B.5 Speech Coding, Multiplexing, and Channel Coding

30.B.5.1 Speech Coder

The speech coder used in UMTS is an *Adaptive Multi Rate* (AMR) coder that also has a strong similarity to the enhanced speech coder used in GSM. AMR codecs are based on *Algebraic Code Excited Linear Prediction* (ACELP) (see Chapter 24). WCDMA-AMR contains eight different encoding modes, with source rates ranging from 4.75 to 12.2 kbit/s, as well as a “background noise” mode.⁴

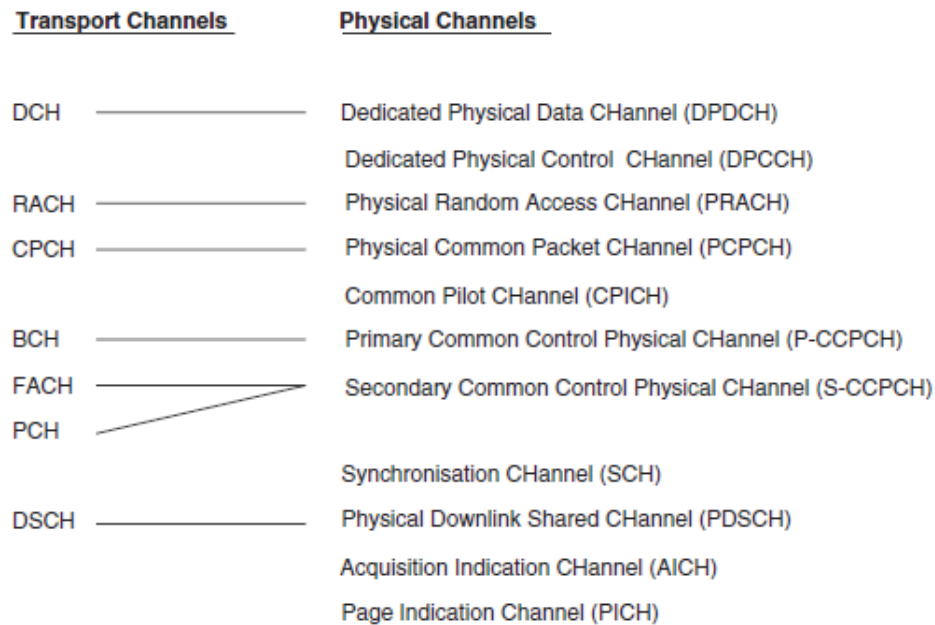


Figure 30.B.6 Matching of physical and logical channels.

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⁴ A later standardized “wideband” AMR that results in even better speech quality uses nine modes with rates between 6.6 and 23.85 kbit/s.

30.B.5.2 Multiplexing and Interleaving

Multiplexing, coding, and interleaving are very complicated procedures that allow a high degree of flexibility. A data stream coming from upper layers has to be processed before it can be transmitted via the transport channels on the air interface. Transport channels are processed in blocks of 10-, 20-, 40- or 80-ms duration. We first discuss multiplexing and coding in the uplink. The block diagram in Figure 30.B.7a illustrates the order of the processes involved in multiplexing and coding:

- When processing a transport block, the first step is to append a *Cyclic Redundancy Check* (CRC) field. This field, which can be 8, 12, 16, or 24 bits long, is used for the purpose of error detection. It is calculated for each block of data for one transmission time interval from the code polynomials:

$$G(D)=D^8+D^7+D^4+D^3+D+1 \quad \text{for 8-bit CRC} \quad (30.B.1)$$

$$G(D)=D^{12}+D^{11}+D^3+D^2+D+1 \quad \text{for 12-bit CRC} \quad (30.B.2)$$

$$G(D)=D^{16}+D^{12}+D^5+1 \quad \text{for 16-bit CRC} \quad (30.B.3)$$

$$G(D)=D^{24}+D^{23}+D^6+D^5+D+1 \quad \text{for 24-bit CRC} \quad (30.B.4)$$

and attached at the end of the block.

- Afterwards the datablocks are concatenated or segmented into blocks that have suitable size for channel coding. The blocks should not be too small, because that increases the relative impact of the overhead (tail bits), and makes the performance of turbo codes worse. On the other hand, the blocks should not be too big; otherwise, decoding can become too complicated. For convolutional encoding the blocksize is typically 500 bits, for turbo codes approximately 5,000 bits. Tail bits are appended to help the decoder: 8 tail bits if convolutional encoding is used, and 4 tail bits for turbocoding.
- The blocks are then encoded with convolutional codes or turbo codes, details of which are discussed in the next subsection.
- The resulting encoded blocks then undergo *radio frame size equalization*. This makes sure that the amount of data is the same for each radio frame.
- In case the block spans more than one frame of length 10 ms, *interframe interleaving* is applied, which interleaves bits over the different frames of this block.
- If necessary, the blocks are then divided into 10-ms transmission blocks, a process called *radio frame segmentation*.
- The encoded block then undergoes *rate matching* – i.e., the rate of the block is then adapted to the desired rate by puncturing or selected bit repetition. Repetition is usually preferred, except for some special high-data-rate cases.
- If multiple data transport channels are transmitted, the resulting transmission blocks or frames are then time multiplexed. Each block is accompanied by a TFCl. This contains the rate information for the current block, and is therefore very important – if the TFCl is lost, then the whole frame is lost.
- The resulting data stream is then fed into a second interleaver, which interleaves the bits over one radio transmission frame, *intraframe interleaving*. In case multiple physical data channels are used, the transmission frames are then mapped to these multiple channels. Otherwise, a single dedicated physical channel is used.

The multiplexing operation in the downlink is slightly different, as outlined in Figure 30.B.7b, as the order of some of the steps is different. The main difference lies in the insertion of DTX (Discontinuous Transmission) bits, which indicate when to turn the transmission off. Depending on whether fixed or variable symbol positions are used, DTX indication bits are inserted at different points in the multiplexing/coding chain.

Coding

There are two modes of channel coding:

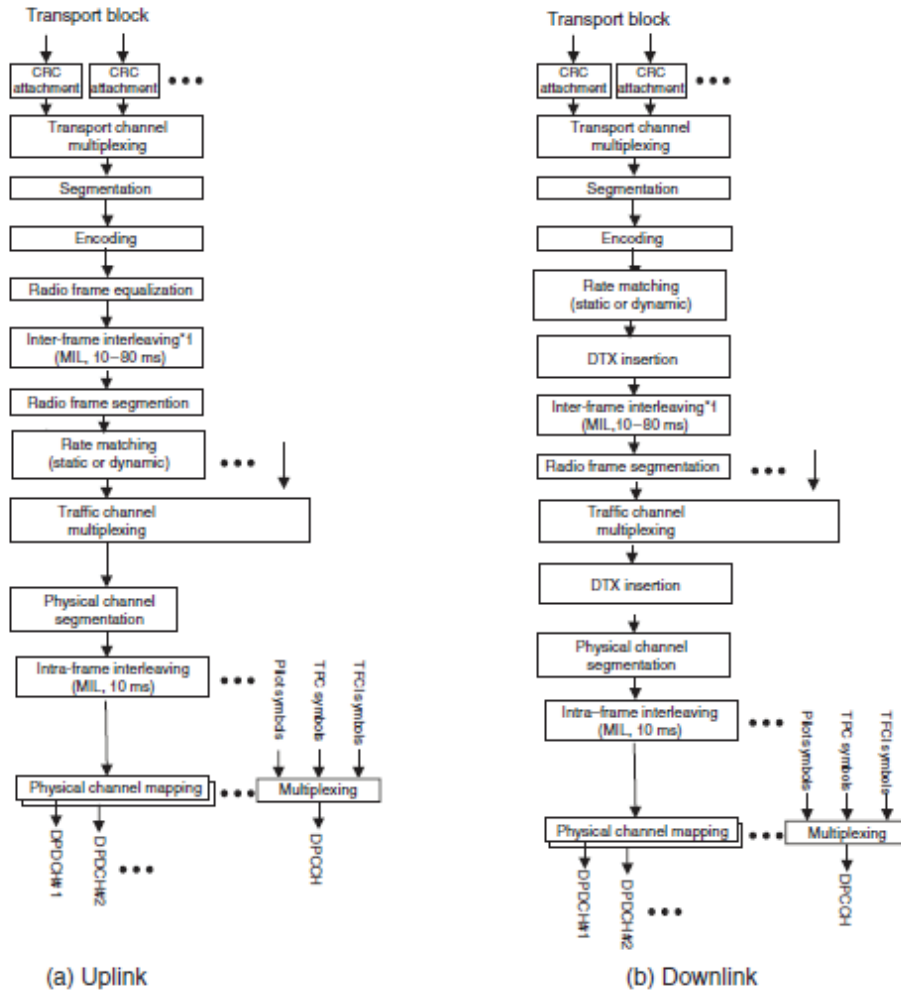


Figure 30.B.7 Multiplexing and coding.

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1. Convolutional codes are used with a coding rate of 1/2 for common channels and 1/3 for dedicated channels. Convolutional codes are mainly used for “normal” applications with a data rate of up to 32 kbit/s. The constraint length of the encoders is 9. Figure 30.B.8 shows the structure of the two encoders. The code polynomials for the rate-1/2 encoder are:

$$G1(D) = 1 + D^2 + D^3 + D^4 + D^8 \quad (30.B.5)$$

$$G2(D) = 1 + D + D^2 + D^3 + D^5 + D^7 + D^8 \quad (30.B.6)$$

and for the rate-1/3 encoder:

$$G1(D)=1+D^2+D^3+D^5+D^6+D^7+D^8 \quad (30.B.7)$$

$$G2(D)=1+D+D^3+D^4+D^7+D^8 \quad (30.B.8)$$

$$G3(D)=1+D+D^2+D^5+D^8 \quad (30.B.9)$$

2. Turbo codes are mainly used for high-data-rate (> 32 kbit/s) applications. Two recursive systematic convolutional encoders are employed (see Figure 30.B.9). The data stream is fed into the first one directly, and into the second one after passing an interleaver. Both encoders have a coding rate of $1/2$. Thus, output is the original bit X or X' and the redundancy bits Y or Y' , which are output from the recursive shift registers. However, as X equals X' only X , Y , and Y' are transmitted. Thus, the code rate of the turbo encoder is $1/3$. For more details on the principles of turbo coding, see Sec. 13.6.

Table 30.B.1 provides an overview of the different coding modes used in different channels.

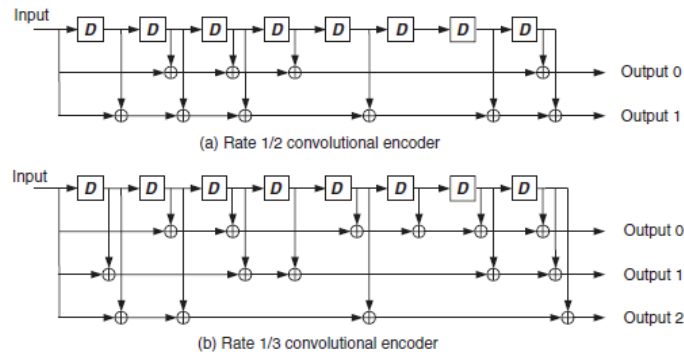


Figure 30.B.8 Structure of convolutional encoders.

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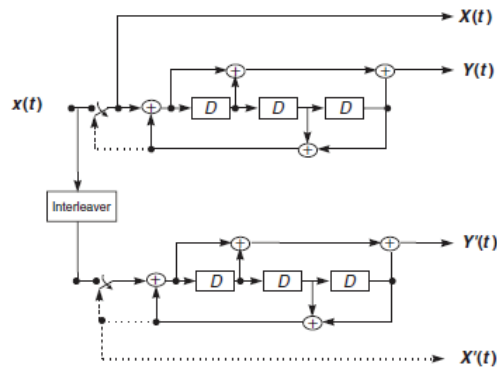


Figure 30.B.9 Structure of a turbo encoder.

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Table 30.B.1 Forward error correction and different logical channels

Transport channel type	Coding scheme	Coding rate
BCH	Convolutional code	1/2
PCH		
RACH		
CPCH, DCH, DSCH, FACH	Turbo code	1/3, 1/2
		1/3

30.B.6 Spreading and Modulation

30.B.6.1 Frame Structure, Spreading Codes, and Walsh–Hadamard Codes

WCDMA relies on CDMA for multiple access. However, transmission timing is still based on a hierarchical timeslot structure similar to GSM's: frames of duration $T_f = 10$ ms are divided into 15 timeslots, each of which has a 12-bit-long *System Frame Number* (SFN). Each timeslot has a duration of 0.667 ms which equals 2,560 chips. The configuration of frames and timeslots is different for uplink and downlink.

WCDMA uses two types of code for spreading and multiple access: *channelization codes* and *scrambling codes*. The former spread the signal by increasing the occupied bandwidth in accordance with the basic principle of CDMA. The latter do not lead to bandwidth expansion but help to distinguish between cells and/or users. In the following we discuss the codes and modulation for the different channels.

Channelization codes in WCDMA are *Orthogonal Variable Spreading Factor* (OVSF) codes (as discussed in Section 19.3.2, see also Section 30.B.5.1).

For scrambling codes, a long and a short code exist. Both are complex codes, and are derived from real-valued codes in accordance with the following expression:

$$C_{\text{Scrambler}}(k) = c_1(k) \cdot (1 + j \cdot (-1)^k \cdot c_2(2 \lfloor k/2 \rfloor)) \quad (30.B.10)$$

Here, k is the chip index and c_1 and c_2 are real-valued codes.

For short codes c_1 and c_2 are two different members of the very large Kasami sets of length 256 (see Section 19.3.2). It is worth noting that the duration of the short code equals symbol duration only for spreading factor 256. Otherwise, a “short” code in WCDMA is not a short code in the sense of Chapter 19.

The long code is a Gold code, a combination of two Pseudo Noise (PN)-sequences that each have length $2^{25} - 1$. The I- and Q-part, c_1 and c_2 in Eq. (30.B.10), are versions of the same Gold sequence, shifted relative to each other. The codes are truncated to a length of 10 ms – i.e., one frame.

30.B.6.2 Uplink

Dedicated Channels

Figure 30.B.10 is a block diagram illustrating spreading and modulation for the uplink.

- Under normal circumstances, user data (DPDCH₁) and control data (DPCCH) are transmitted on the in-phase component and the control channel on the quadrature-phase component. First, channelization codes c_d and c_c are applied to the data and the control channel, respectively. As mentioned above these codes actually increase the signal bandwidth. Afterwards, the I- and Q-branch are treated as a complex signal. Processing the control and data channel like this is called I–Q multiplexing.

- If the data rate for the user is very high, then up to five additional data channels may be transmitted in parallel on the I-branch and Q-branch. These channels are then distinguished by applying different spreading codes, $c_{d,k} \ k \in [1, \dots, 3]$ (!). The spreading factor is in this case 4. Therefore, the total data rate is given by:

$$\frac{4 \text{ Mchip/s}}{4 \text{ chips/bit}} \cdot 6 \cdot \frac{1}{3} \quad (30.B.11)$$

The 4 chips/bit represent the spreading. The factor 6 relates to the maximal six transmitted channels and factor 1/3 relates to channel coding. Thus, the maximal achievable net user data rate is 2 Mbit/s.

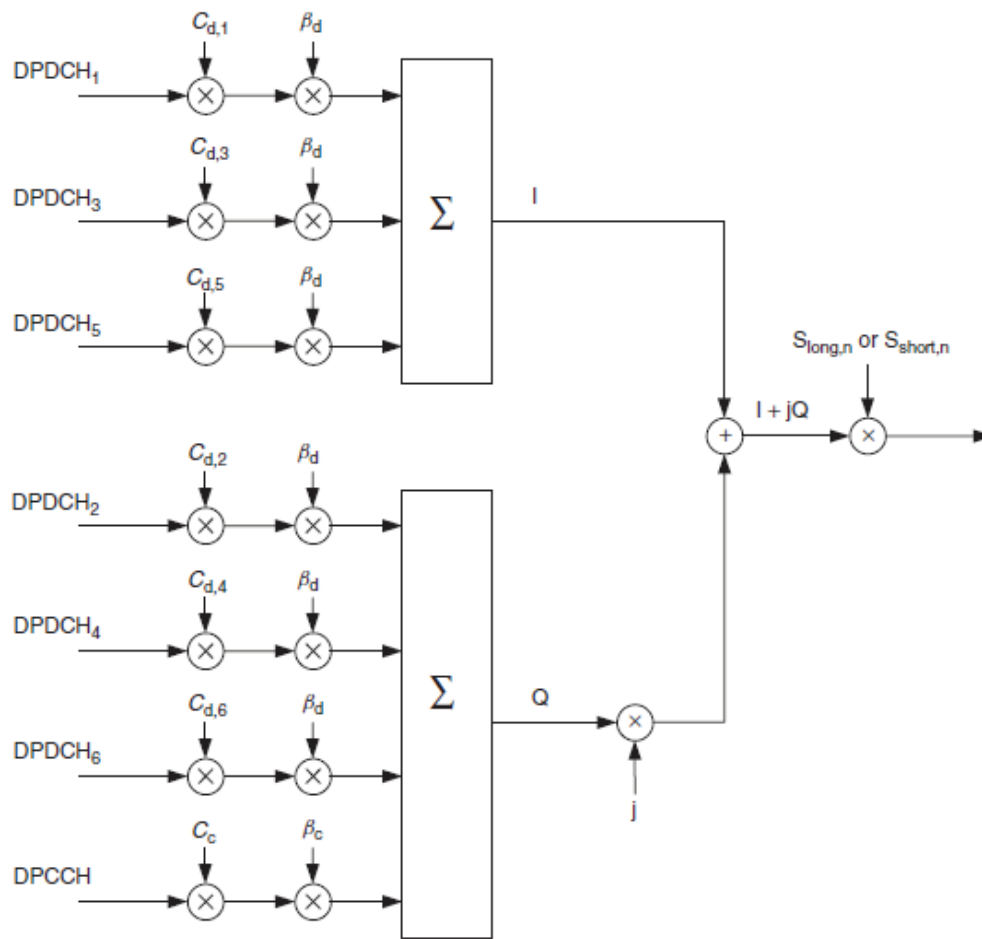


Figure 30.B.10 Spreading and modulation on the uplink.

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- The transmit power of the control channel relative to the data channel(s) is determined by the ratio of spreading factors, β_c/β_d . The received noise power is proportional to the (unspread) bandwidth and thus the data rate. Consequently, the transmit power has to be the higher the lower the spreading factor in order to make sure that the Signal-to-Noise Ratio (SNR) at the receiver is the same for different data. The spreading factor of control data is always 256.
- Then the complex scrambling code $S_{\text{long},n}$ or $S_{\text{short},n}$ is applied and the signal is fed into the complex modulator. The complex signal usually shows a power imbalance between the I- and Q-factor. In order to deal with this problem, spreading and scrambling codes are designed in such a way that the signal constellation is rotated by 90° between two subsequent chips.
- The resulting signal is modulated using Quadrature Amplitude Modulation (QAM) with Nyquist basis pulses. More precisely, the basic pulses are raised cosine pulses with a roll-off factor of $\alpha = 0.22$. Modulation precision is supposed to be better than 17%. Precision is measured by taking the ratio of the power of the difference signal between the actual and the ideal signal and the power of the ideal signal – *Error Vector Magnitude* (EVM).

I-Q multiplexing is used in the uplink in order to limit the crest factor. Even when there are no user data to be transmitted – e.g., in no-voice periods – the control channel is continuously active. Therefore, the RF amplifier in the UE does not have to switch on and off. This eases hardware requirements as the amplifier has to cover a smaller dynamic range. Furthermore, amplitude variations from slot to slot or burst to burst could lead to interference with audioprocessing equipment, like the microphone of the UE or the hearing aid of the user, as these activities would be in the audible frequency range.

We next turn to look in more detail at the control channel. It carries the pilot bits, the TPC bits, the FBI, and the TFCl. As the spreading factor is constantly 256, 10 bits of control information are transmitted per timeslot, see Figure 30.B.11.

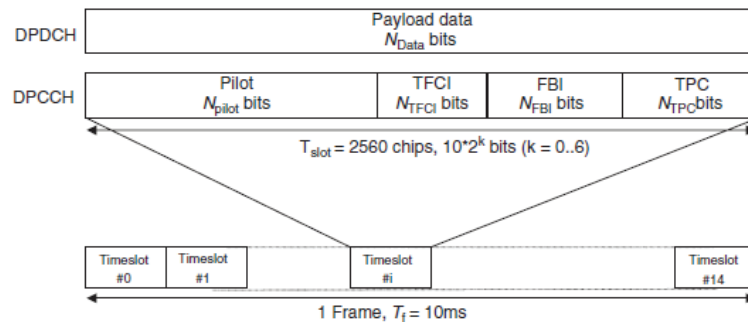


Figure 30.B.11 Frame structure on the uplink.

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Random Access Channel

Consider now the burst structure of the RACH (see Figure 30.B.12). The start time for RACH transmission should be $t_0 + k \cdot 1.33 \text{ ms}$, where t_0 refers to the start of a regular frame on the BCH. At the beginning of an access burst several preambles of length 4,096 chips are transmitted. The first preamble has power that is determined by “open-loop” power control – i.e., the UE measures the strength of the BCH, and from that concludes the amount of small-scale-averaged path gain; this (plus a safety margin) in turn determines transmit power. If the UE does not receive an acknowledgement of its access request on the AICH, it transmits the preamble again, with increased power. After receiving an acknowledgement, the actual access data (a field that is either 10 ms or 20 ms long) are transmitted.

The preamble contains one of a set of 16 predefined signature sequences, which are transmitted with a spreading factor of 256. The “data” containing message part of length 10 ms is similar to the DPCH divided into 15 timeslots. Within each of those timeslots 8 pilot bits and 2 TFCI bits are transmitted with a spreading factor of 256; this constitutes the control information for layer 1. Furthermore, a “data” message, usually containing control information for layer 2, is transmitted with a spreading factor between 32 and 256. Layer-1 control information and the “data” message are transmitted simultaneously by I–Q multiplexing (see Figure 30.B.13). Thus, the structure of the message part of the RACH is very similar to a frame of a dedicated physical channel. However, neither a TPC nor an FBI field are transmitted.

Base Station Processing

The BS has to process the received signal periodically in the following way [Holma and Toskala 2000]:

- It receives one frame, despreads it, and stores it with the sampling frequency given by the highest used data rate of this frame. Note that different data rates and spreading factors may be used during one frame.
- For each timeslot: (i) the channel impulse response is estimated using the pilot signals; (ii) the Signal-to-Interference Ratio (SIR) is estimated; (iii) power control information is transmitted to the UE; and (iv) power control information from the UE is decoded and transmission power is adapted accordingly.
- For every second or fourth timeslot: FBI bits are decoded. Two or four FBI blocks make up one complete FBI command, like an update of antenna weights.
- For each 10-ms frame: TFCI information is decoded to get the decoding parameters for the DPDCH.
- For each interleaver period, which is either 10, 20, 40, or 80 ms: user data transmitted via the DPDCH is decoded.

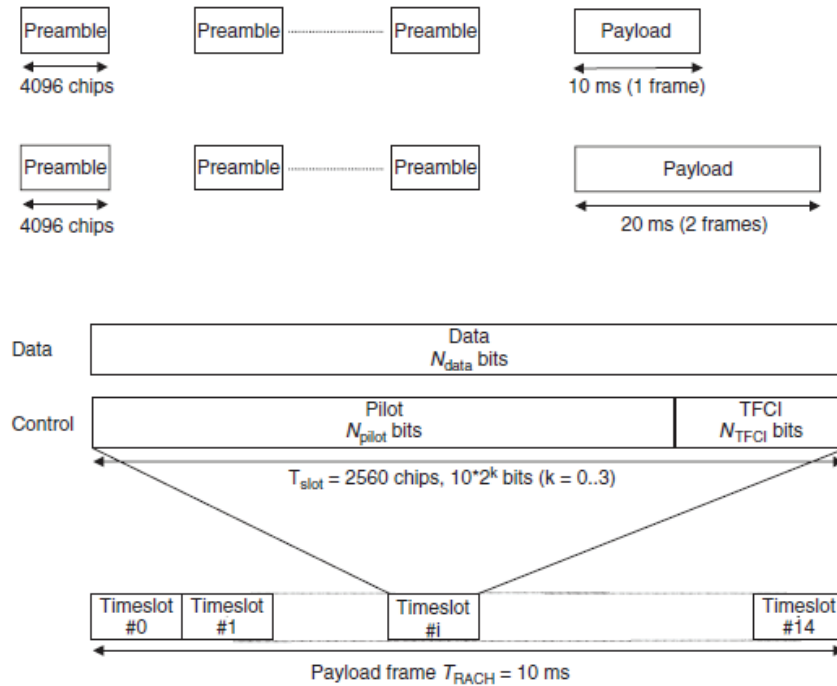


Figure 30.B.12 Structure of random access transmission.

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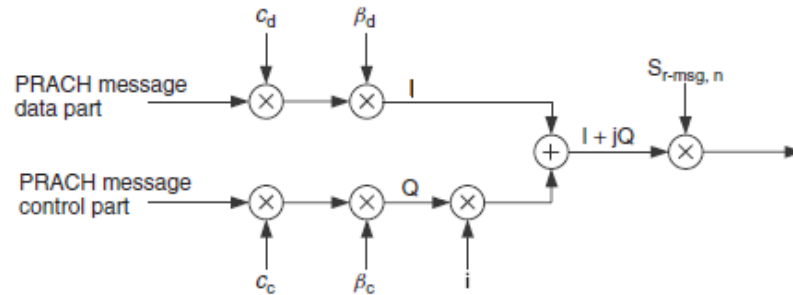


Figure 30.B.13 Modulation on the physical random access channel.

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Spreading Codes

The uplink uses OVFS codes for spreading. However, they are not used for channelization (distinguishing between users in the uplink). Therefore, different users can use the same spreading codes. As a matter of fact, assignment of spreading codes to the different channels of any UE is predefined. The DPCH is always spread with the first code, the code with index 0, using spreading factor 256. A DPDCH channel is spread using the code with index $SF/4$, where SF is the spreading factor of the channel. In case multiple data channels are transmitted, the spreading factor is 4, when spreading codes with indices 1, 2, or 3 are used; one code is used for both a channel on the I-branch and a channel on the Q-branch. The spreading factor of a traffic channel may vary from frame to frame. There are additional code selection criteria for the PRACH and PCPCH.

Scrambling Codes

As mentioned above, signals from different users are distinguished by different scrambling codes. Both “short” and “long” codes may be used (see Chapter 19). There are no strict rules about when to apply short or long codes. However, it is recommended to use short codes when the BS has multiuser detection capability (see Section 28.1). Short codes limit the computational complexity for multiuser detection, as the size of the cross-correlation matrices involved is smaller. In case no multiuser detection is implemented, long codes should be used as they provide better averaging of interference than short codes.

The BS assigns the specific scrambling code to the UE during connection setup with the *Access Grant Message*.

Codes for Random Access Channels

Random access channels use codes that are different from those in regular channels. These codes are specific to one BS, and two neighboring BSs should not use the same or similar codes. The preamble is transmitted first, which serves for synchronization and identification (see below). The preamble is designed in such a manner that it is very robust against initial uncertainties in frequency synchronization. Furthermore, RACH codes are only transmitted with a binary alphabet, which eases receiver design.

30.B.6.3 Downlink

Spreading and modulation of the data and control channels for the downlink is different from the uplink. In the downlink, data and control channels are *time-multiplexed* and the resulting single bitstream is then Quadrature-Phase Shift Keying (QPSK) modulated. We may interpret QPSK modulation of a single bitstream as a serial/parallel conversion into two streams on the I- and Q-branch. The resulting I-signal and Q-signal are

spread separately using the same channelization code. The complex scrambling code is then applied to the resulting complex signal. Finally, the scrambled complex signal is fed into the complex modulator.

Figure 30.B.14 illustrates the frame and timeslot structure for the DDPCH. The first datablock containing bits is transmitted. Then the TPC field is sent, followed by the TFCI field. A second datafield is transmitted thereafter. The timeslot is concluded with the pilot field. A spreading factor between 4 and 512 can be used. The purpose and properties of the pilot, TFCI, and TPC fields are the same as for the uplink.

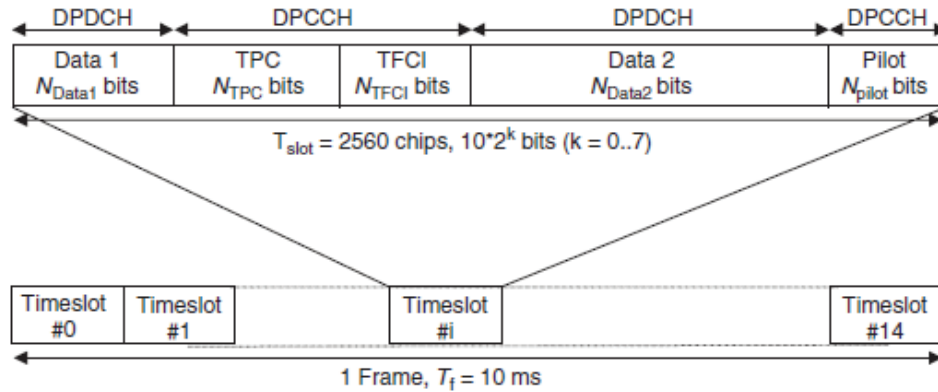


Figure 30.B.14 Frame and timeslot structure on the downlink.

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The spreading factor for one user does not vary from frame to frame. It is chosen to accommodate the highest occurring data rate of this user; if a lower rate is required instantaneously, DTX is employed – i.e., transmission is blanked out for a while. The UE can be informed about the current data rate by the TFCI field.

The reasons that DTX can be employed in the downlink (in contrast to the uplink) are that (i) the appearance of audible interference resulting from on/off switching of transmit amplifiers is not an issue on the downlink; the common control and synchronization channels, like the BCH or the SCH, are transmitted continuously anyway; and (ii) high values of the crest factor cannot be avoided anyway as transmission of several CDMA signals in parallel always leads to high crest factors.

UE Processing

The UE receiver has to perform operations for the downlink similar to those of the BS receiver in the uplink. However, there are some differences [Holma and Toskala 2000]:

- The spreading factor of all channels apart of the DSCH is constant over time.
- There is no need for the FBI field.
- There is one CPICH in addition to the pilots of dedicated physical channels, which improves the channel estimate.
- Smart antennas can be used in the downlink.

Channelization Codes

The signals for different users are inherently synchronized in the downlink, as they come from the same transmitter – i.e., the BS. Therefore, OVFS codes enable good separation of the signals for different users within one cell. The BS informs the UE about the code used during connection setup. The BS ensures that the same code (or a “mother code” in the OVFS tree) is only used once in the cell. If necessary, the code can be changed during a connection.

Scrambling Codes

Long codes for the downlink are the same as in the uplink; short codes are not used for the downlink. There are 512 primary scrambling codes,⁵ which are divided into 64 groups of 8 codes each. Primary, secondary, and alternative codes are unambiguously related to each other. Each cell has exactly one primary scrambling code that is used on the CCPCH and CPICH. The other downlink channels might use the same primary or a related secondary code. However, use of more than one scrambling code only makes sense when other measures, like smart antennas, are used to increase cell capacity. Otherwise, one scrambling code in conjunction with spreading codes already achieves maximum user separation.

Overview

Table 30.B.2 provides an overview of the different codes used.

Table 30.B.2 Overview of different channelization and scrambling codes

	Channelization codes (variable chip rate)		Scrambling codes (constant chip rate)		
	Uplink	Downlink	Uplink	Downlink	
Separates	Channels (I/Q, DPDCH) of one user	Channels (DPDCH) and users	Users	Cells	
	Assigned to connection	Assigned to connection and user	Assigned to user by BS	Assigned to cell	
Reuse	Within the same cell	In all other cells	In other cells	In distant cells (code planning)	
Selection	Fixed (given by the SF)	Variable	Variable	Fixed	
Length of code	Short	Short	“Short”	Long	Long
Enables	Variable data rates	Different data rates	Multisuser detection		Cell search
Code family	OVSF (real)	OVSF (real)	Complex, based on VL Kasami	Complex, based on segments of Gold codes	Complex, based on segments of Gold codes
Codelength	4–256	4–512	256	38,400	38,400
Number of codes	Up to 256		$> 10^6$	$\gg$	512 (8, 192)

High-Data-Rate Users

There are basically two possibilities to support high data rates in UMTS:

1. Transmission with a low spreading factor. This is the most straightforward way to increase the data rate. The lowest admissible spreading factor in WCDMA is 4; note that a spreading factor of 1 would lead to a pure Frequency Division Multiple Access (FDMA) system. The drawback of a low spreading factor is that InterSymbol Interference (ISI) becomes significant: with a spreading factor of 4, the symbol duration can be as low as 1 μ s. Thus, the receiver has to cope not only with interchip interference, which is effectively dealt with by a Rake receiver structure, but also with severe ISI that needs to be eliminated by an equalizer (Chapter 14). This additional implementation complexity may increase the cost of a UE.

⁵ Additionally, there are 512×25 secondary scrambling codes and 8,192 left and 8,192 right alternative scrambling codes for compressed mode.

2. The data stream can be serial/parallel converted and then transmitted using multiple codes, where each code has a sufficiently high spreading factor. This is called “multicode transmission.” ISI is not a problem with this approach. However, two other problems appear: (i) the crest factor of a multicode signal is high (this is not a problem on the downlink, as multiple users have to be supported anyway); and (ii) efficiency is lowered as every additionally transmitted code leads to additional overhead.

30.B.7 Physical-Layer Procedures

30.B.7.1 Cell Search and Synchronization

The search for the signal of the strongest BS and synchronization with this BS has to be done in several steps.

In a first step, the UE *synchronizes with the timeslot* timing of the strongest signal it can observe. This is done by searching for the 256-chip primary synchronization code. Every BS periodically transmits this sequence, and it is identical for the whole system. The UE thus just correlates the incoming signal with this synchronization sequence to identify all available BSs. The output of this correlator shows several peaks at different delays, corresponding to the signals of different BSs and the various echoes (MPCs) of these signals. By selecting the highest peak, the UE achieves timeslot synchronization with the strongest signal. In other words, the UE is aware of the commencement of a timeslot in the strongest BS. However, it still does not know about frame timing, as it does not know, e.g., whether the first or the tenth timeslot is currently received.

Therefore, *frame synchronization* has to take place in the next step. This is done by observing the secondary synchronization channel; in the same step the codegroup of the primary scrambling code used by the BS is determined. As discussed earlier, the CCPCH and CPICH are transmitted in the downlink with one of 512 primary scrambling codes. These 512 primary codes are divided into 64 groups. The secondary synchronization channel transmits a sequence of 15 codes in 15 consecutive timeslots. Each code is a letter of the same alphabet. Thus, we may interpret the code sequence as a word with 15 letters. This codeword is repeated periodically. The UE can thus determine the group used and achieve frame synchronization by observing this codeword. Codewords are chosen in such a manner that a cyclic shift of one codeword never results in another codeword. Therefore, the UE can determine which codeword is transmitted (this indicates the used codegroup) and where the codeword starts (this indicates frame timing).

Let us illustrate this procedure with a simplified example. Assume, that the alphabet size used is five and three codewords of length 4 can be transmitted: (i) *abcd*; (ii) *aedc*; and (iii) *bcde*. Further, assume that *edca* is received. As (i) does not feature the letter *e* and (iii) does not feature the letter *a*, the received word has to be a cyclically shifted version of (ii). Once we know that (ii) is transmitted, we can easily see that frame timing has to be adjusted by one slot.

After the codegroup has been determined, the UE has to determine which code of this group is used on the CCPCH. Therefore, the CCPCH undertakes symbolwise correlation with all possible eight codes of that codegroup. Once this is achieved, the CCPCH can be properly demodulated.

30.B.7.2 Establishing a Connection

A connection initialized by the UE requires the following procedures:

1. The UE synchronizes itself with the strongest BS as described in the previous subsection.
2. The UE decodes the BCH and thereby acquires information regarding: (i) the spreading code of the preamble and the scrambling code for the message part of the PRACH; (ii) the codes available; (iii) the access timeslots available; (iv) possible spreading factors for messages; (v) the interference level on the uplink; and (vi) the transmit power of the CCPCH.
3. The UE selects a spreading code for the preamble and a scrambling code for messages.
4. The UE determines the spreading factor for the message.
5. Based on the measured signal strength of the CCPCH and information on the transmit power of the BS, the UE estimates the attenuation on the uplink. Based on the attenuation estimate and information regarding the uplink interference level, the UE estimates the necessary transmission power for the preamble.
6. The UE randomly selects an access timeslot and a signature from the available set.

7. The UE transmits the preamble. In case of successful acquisition, the BS transmits an acknowledgement on the AICH.
8. In case the UE does not receive an acknowledgement from the BS it repeats the preamble with increased transmission power.
9. Once the BS indicates acquisition of the preamble, the UE starts to transmit the access message in the next available timeslot.
10. The UE waits for an access grant message from the network. If it does not receive this message within a predetermined time, it repeats the steps from step 5 on.

30.B.7.3 Power Control

Power control is an essential part of a CDMA system, as it is necessary to control mutual interference (see Sec. 19.3.3). Inner-loop power control in WCDMA, in particular, is supposed to adapt to *smallscale* fading for speeds up to 500 km/h. Therefore, the power control procedures have to be rather fast. An update of transmit power occurs with every timeslot – i.e., every 0.667 ms. There is also outer-loop power control which continuously adjusts the target SIR for inner-loop power control.

Uplink

The uplink uses a closed-loop procedure for power control. The BS estimates the power of the received signal and controls it by transmitting TPC instructions to the UE, which changes its transmit power accordingly. TPC bits are transmitted with the DPCCH, and contain instructions to increase or decrease power. The possible stepsizes are 1 dB or 2 dB with uncertainties of ± 0.5 dB or 1 dB. However, transmit power is not decreased below, for example, -44 dBm. Even so, given a maximal transmission power of 33 dBm for class-1 units, the RF amplifier has to cover a dynamic range of 77 dB, which is a rather severe hardware requirement.

As for soft handover, the situation is rather complicated as the UE can receive different TPC bits from the different BSs involved. The UMTS standard specifies an algorithm that determines a means of obtaining a combined power control command from these different commands. The combined instruction is a function of weighted single-power-control commands. The weighting applied is proportional to reliability of the individual signals. Some particular rules apply for special situations. For example, for both compressed mode and the common packet channel a special procedure is implemented, as longer gaps between transmissions and TPC commands occur.

It is not possible to use closed-loop power control on the PRACH, as a dedicated channel has not yet been established between the UE and the BS. Therefore, open-loop power control has to be employed. The UE measures the average received power of the CCPCH over some time to average out small-scale fading effects. This is necessary because, due to the frequency duplex, it is not possible to determine from the instantaneous received power on the downlink, the level of instantaneous received power on the uplink (see Sec. 16.1.7). Average received power allows estimation of just the necessary *average* transmit power for the uplink. However, this is at least a good starting value.

Downlink

All signals on the downlink (to one UE) suffer from the same attenuation. Furthermore, downlink signals are orthogonal, due to the use of OVSF codes for channelization, so that the despreading operation results in a good SIR.⁶ Therefore, power control on the downlink aims mainly at maintaining a good SNR.⁷ A closed-loop is used for power control in the downlink. Each UE measures signal strength and quality and transmits TPC commands to the BS via the DPCCH. As long as at least one UE requests more transmission power the BS increases the power of all channels. Therefore, UEs at cell borders actually control the transmission power of the BS.

One problem is that the UE estimates the SNR *after* the Rake receiver. A low-performance device, which employs only a few Rake fingers, captures less received signal energy than a more sophisticated device with

⁶ Note, however, that orthogonality can be destroyed in frequency-selective channels.

⁷ This argumentation only considers one cell. Considering intercell interference, the SIR in one cell might be decreased by increasing the transmission power of all channels. However, this in turn decreases the SIR in all neighboring cells.

more Rake fingers. Therefore, a low-performance device tends to request more transmit power from the BS than an advanced one. Thus, the UMTS network levels out differences in the designs of units by ensuring good receive qualities independent of receiver design quality. Of course, this might limit the motivation of manufacturers to develop and produce good UEs.

Transmit Diversity

The downlink may implement the option of transmit diversity with two antennas. It distinguishes between closed-loop diversity, for which the transmitter requires Channel State Information (CSI), and open-loop diversity, for which the transmitter does not require any information about the channel. Three modes of diversity are specified:

- Orthogonal space–time block coding with two antennas – i.e., the Alamouti code (see Section 16.1.5).
- The transmit signal might be switched from one antenna to the other. This mode is only used for the SCH.
- Closed-loop transmit diversity: two antennas can transmit the same stream, by applying complex weights for each antenna. The pilot bits on signals from the two antennas are mutually orthogonal sequences. Therefore, the receiver can estimate the channel impulse responses of the two channels separately. Antenna weights are then determined such that signals from the two transmit antennas add up constructively at the UE (see Sec. 16.1). The computed weights are digitized, and transmitted to the BS via the FBI field of the DPCCH. This mode of diversity is optional for BSs, but UEs have to be able to support it. Support for more advanced multiple-antenna techniques was introduced in later releases.

30.B.7.4 Handover

Intrafrequency Handover

A connection handover between two BSs on the same carrier frequency is performed as a *soft handover*. The UE has a connection to both BSs during handover (see Section 21.6). Thus, signals from both BSs are used during this time and the Rake receiver processes them similarly to two paths of a multipath signal with two or more fingers. As BSs use different scrambling codes in WCDMA, the Rake receiver in the UE has to be able to apply different codes in each finger.

In preparation for soft handover, the UE has to acquire synchronization with other BSs. This synchronization process is similar to the one described above, apart from the fact that the UE has a priority list for codegroups. This list contains the codegroups used by the neighboring cells for handover, and is continuously updated.

Cell selection for soft handover can be made by different criteria, like signal strength after despreading or wideband power (Received Signal Strength Indication RSSI). No particular algorithms are specified in the standard. However, it is suggested to divide cells into the active cell and neighboring candidate cells that provide good signal strength.

A so-called *softer handover* is a special case of soft handover, in which the UE switches between two sectors of the same BS. Algorithms and processes are similar to those used for soft handover with the difference that only one BS is involved.

Interfrequency Handover

This kind of handover takes place in the following situations:

- Two BSs employ different carriers.
- The UE switches between hierarchy layers in the hierarchical cell structure.
- Handover to other providers or systems – e.g., GSM – is necessary.
- The UE switches from TDD mode to FDD mode or vice versa.

Interfrequency handover is a “hard” handover during which the connection between the UE and the old BS is first interrupted before the UE establishes a new connection with a new BS. There are two ways of measuring

signal strength, etc. on other frequencies while the old connection is still active: (i) the UE might feature two receivers, so that one can measure on other frequencies while the first one still receives user data; and (ii) transmission is done in compressed mode, so that data that normally are transmitted within a 10-ms frame are compressed to 5 ms. The time that is freed up can then be used for measurements on other frequencies. Compression can be achieved, for example, by puncturing the data stream or reduction of the spreading factor.

30.B.7.5 Overload Control

The UTRAN has to ensure that it does not accept too many users or provide users with too high data rates, as then the system would be overloaded. Several means can be used for this [Holma and Toskala 2000]:

- No increase in transmit power in the downlink, even when one UE requests it. This of course implies lower signal quality in the downlink. If data packets are transmitted, the Automatic Repeat reQuest (ARQ) rate increases, which in turn lowers throughput. In case of voice transmission, audio quality is decreased.
- The target Signal-to-Interference-and-Noise Ratio (SINR) in the uplink can be decreased. This decreases the transmission power of UEs. It also results in decreased transmission quality for all subscribers.
- Throughput of the data packet channels can be lowered. This decreases the speed of data connections but maintains the quality of voice services.
- The output data rate of UMTS voice encoders can be decreased, as UMTS employs an encoder with a variable rate. This decreases audio quality, but has no impact on data applications.
- Some active links can be transferred to other frequencies.
- Regular phone calls can be handed over to the GSM network.
- Data connections or phone calls can be terminated.

30.B.8 Glossary for WCDMA

3GPP	Third Generation Partnership Project
ACELP	Algebraic Code Excited Linear Prediction
ACLR	Adjacent Channel Leakage Ratio
AD	Access Domain
AICH	Acquisition Indication Channel
AMR	Adaptive Multi Rate
AN	Access Network
ARQ	Automatic Repeat reQuest
ATM	Asynchronous Transfer Mode
BCH	Broadcast CHannel
CCPCH	Common Control Physical Channel
CCTrCH	Coded Composite Traffic CHannel
CDMA	Code Division Multiple Access
CN	Core Network
CND	Core Network Domain
CPCH	Common Packet CHannel
CPICH	Common Pilot CHannel
CRC	Cyclic Redundancy Check
DCH	Dedicated (transport) Channel
DPCCH	Dedicated Physical Control CHannel
DPDCH	Dedicated Physical Data CHannel
DSCH	Downlink Shared Channel
EVM	Error Vector Measurement
FACH	Forward Access CHannel
FBI	Feed Back Information
IMT	International Mobile Telecommunications
IP	Internet Protocol
ISI	Inter Symbol Interference

IWU	Inter Working Unit
JD-TCDMA	Joint Detection-Time/Code Division Multiple Access
ME	Mobile Equipment
MSS	Mobile Satellite Service
MT	Mobile Termination
Node-B	Base station
PCH	Paging Channel
P-CCPCH	Primary Common Control Physical CHannel
PCPCH	Physical Common Packet Channel
PDSCH	Physical Downlink Shared CHannel
PICH	Page Indication Channel
PRACH	Physical Random Access CHannel
QoS	Quality of Service
RACH	Random Access CHannel
RAN	Radio Access Network
RNC	Radio Network Controller
RNS	Radio Network Subsystem
SAP	Service Access Point
SAPI	Service Access Point Identifier
S-CCPCH	Secondary Common Control Physical CHannel
SCH	Synchronization CHannel
SFN	System Frame Number
SMS	Short Message Service
TA	Terminal Adapter
TE	Terminal Equipment
TFCI	Transmit Format Combination Indicator
TFI	Transport Format Indicator
TPC	Transmit Power Control
UARFCN	UTRA absolute radio frequency channel number
UE	User Equipment
UED	User Equipment Domain
UE-ID	User Equipment in-band IDentification
UMTS	Universal Mobile Telecommunications System
UCPCH	Uplink Common Packet CHannel
USIM	User Service Identity Module
UTRA	UMTS Terrestrial Radio Access
UTRAN	UMTS Terrestrial Radio Access Network
W-CDMA	Wideband CDMA

Further Reading

The most authoritative source for UMTS is, of course, the standard itself, whose most recent version can be found at www.3gpp.org. However, this material is exceedingly difficult to read. Good summaries can be found in [Holma and Toskala 2007] and [Richardson 2005], as well as a number of other monographs. The High-Speed Packet Access (HSPA) part is described very nicely in [Dahlman et al. 2008].

For updates and errata for this Appendix, see wides.usc.edu/teaching/textbook